

AMDERMA, Russian Federation

WMO: 230220

Lat: 69.750N

Lon: 61.700E

Elev: 161

StdP: 14.61

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-28.1	-24.1	-32.5	0.9	-27.7	-29.0	1.1	-23.9	43.6	6.1	39.6	5.2	9.3	90	1.037

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	68.7	61.0	63.3	57.8	58.3	54.5	61.6	67.6	58.0	62.7	54.6	58.3	12.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.9	74.8	65.2	55.5	65.9	60.6	52.0	58.0	57.0	27.7	67.4	25.2	62.9	23.0	58.5	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 36.6	32.5	29.0	-30.9	77.5	6.3	5.4	-35.4	81.4	-39.1	84.6	-42.6	87.7	-47.2	91.6		
(5)			-31.3	66.4	6.0	3.8	-35.6	69.1	-39.1	71.3	-42.5	73.4	-46.8	76.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	22.4	1.0	0.5	5.7	13.9	24.8	37.3	46.1	46.0	39.7	28.8	15.2	8.2		
	DBStd	19.61	13.84	15.00	13.80	11.53	8.31	8.01	8.78	5.64	5.57	8.19	11.00	14.02		
	HDD50	10188	1519	1386	1372	1082	780	397	188	155	312	656	1044	1297		
	HDD65	15550	1984	1806	1837	1532	1245	831	590	590	758	1121	1494	1762		
	CDD50	117	0	0	0	0	0	16	67	30	4	0	0	0		
	CDD65	4	0	0	0	0	0	0	4	0	0	0	0	0		
	CDH74	37	0	0	0	0	0	6	30	1	0	0	0	0		
	CDH80	4	0	0	0	0	0	0	4	0	0	0	0	0		
	WSAvg	14.0	16.4	16.2	14.8	14.4	13.0	11.7	10.6	11.4	12.3	14.3	16.1	16.8		
	PrecAvg	15.7	1.0	0.7	0.7	0.7	1.0	1.4	2.0	1.9	1.8	1.8	1.3	1.1		
(15) Precipitation	PrecMax	20.0	2.4	2.0	2.5	1.5	1.9	3.0	4.3	3.6	4.8	3.0	3.4	2.1		
	PrecMin	10.0	0.0	0.0	0.2	0.0	0.2	0.4	0.3	0.7	0.7	0.7	0.3	0.2		
	PrecStd	2.4	0.7	0.5	0.4	0.4	0.5	0.7	1.3	0.7	1.0	0.6	0.7	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.9	30.5	33.2	37.4	52.1	73.3	78.0	68.9	59.9	46.3	34.4	32.5		
		MCWB	30.3	29.9	31.9	35.3	46.9	60.6	65.8	61.0	54.2	44.5	33.6	32.1		
	2%	DB	28.3	27.1	30.5	34.2	42.2	63.5	71.5	63.5	53.9	43.1	32.8	31.0		
		MCWB	27.7	26.6	29.9	33.0	39.8	56.5	63.1	58.2	50.6	41.9	32.2	30.4		
	5%	DB	24.7	24.6	28.5	32.3	38.0	56.0	66.3	59.2	50.5	40.6	31.4	29.0		
		MCWB	24.1	24.0	27.9	31.5	36.0	51.3	59.9	55.4	48.6	39.6	30.7	28.4		
	10%	DB	20.7	21.0	25.7	30.2	35.0	49.4	60.9	55.2	47.9	38.4	29.2	26.9		
		MCWB	20.0	20.3	25.0	29.5	33.7	46.2	56.5	52.7	46.2	37.4	28.4	26.3		
	0.4%	WB	30.3	30.0	32.2	35.5	47.4	61.2	67.4	62.4	55.5	44.9	33.9	32.3		
		MCDB	30.5	30.3	32.9	37.0	51.7	70.1	75.8	67.0	59.3	45.7	34.1	32.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	27.7	26.7	30.1	33.2	40.1	56.8	64.1	58.5	51.1	42.1	32.5	30.6		
		MCDB	28.2	27.1	30.4	34.0	42.1	63.1	70.1	62.8	53.1	43.0	32.8	30.9		
	5%	WB	24.0	23.9	28.0	31.6	36.1	51.2	60.4	55.7	48.7	39.6	30.8	28.5		
		MCDB	24.6	24.4	28.5	32.1	37.5	55.9	65.8	59.0	50.4	40.4	31.4	28.9		
	10%	WB	20.0	20.4	25.1	29.5	33.8	46.3	56.5	52.7	46.4	37.6	28.5	26.3		
		MCDB	20.6	21.0	25.7	30.2	34.8	49.3	60.6	55.1	47.8	38.3	29.2	26.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.0	10.7	10.9	11.1	8.0	8.8	10.8	8.4	7.3	6.9	9.7	10.7		
		MCDBR	12.6	11.0	8.8	10.3	11.5	21.9	21.6	17.0	11.8	6.9	8.9	9.5		
		MCWBR	12.8	10.9	8.9	9.9	9.5	15.3	14.3	12.2	9.2	6.5	9.2	9.6		
		MCDBR	12.2	10.7	9.0	9.8	10.9	21.4	20.5	15.9	11.0	6.5	9.0	9.6		
	5% WB	MCWBR	12.3	10.8	9.0	9.7	9.3	15.5	14.4	12.1	8.9	6.3	9.4	9.8		
(40) Clear-Sky Solar Irradiance	taub		0.210	0.241	0.292	0.319	0.334	0.339	0.372	0.353	0.315	0.258	N/A	N/A		
	taud		1.874	1.965	2.036	2.061	2.089	2.289	2.362	2.428	2.457	2.164	N/A	N/A		
	Ebn at Noon		3	147	219	254	265	270	255	246	224	144	0	0		
	Edh at Noon		1	14	27	36	40	34	31	26	19	11	0	0		
(44) All-Sky Solar Radiation	RadAvg		3	95	558	1219	1733	1724	1558	958	495	127	0	0		
	RadStd		0	21	65	99	103	175	212	93	42	18	0	0		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+3.10	+5.00	+6.05	+2.95	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	+3.10	+5.00	+6.05	+2.95	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page